Context Free Grammars

Outline

- Removing Λ-Productions
- Removing Unit-Productions
- Removing Useless Variables
- Normal Form of Context Free Grammars
  - Chomsky Normal Form (CNF)

Killing Λ-Productions

Λ-Productions:

In a given CFG, we call a non-terminal N null able if there is a production N → Λ, or there is a derivation that starts at N and lead to a Λ.

N → …… → Λ

- Λ-Productions are undesirable.
- We can replace Λ-production with appropriate non-Λ productions.

Theorem 23

- If L is CFL generated by a CFG having Λ-productions, then there is a different CFG that has no Λ-production and still generates either the whole language L (if L does not include Λ) or else generate the language of all the words in L other than Λ.
- Replacement Rule.
    1. Delete all Λ-Productions.
    2. Add the following productions:
- For every production of the X → old string

  Add new production of the form X → .., where right side will account for every modification of the old string that can be formed by deleting all possible subsets of null-able Non-Terminals, except that we do not allow X → Λ, to be formed if all the character in old string are null-able

Example

Consider the CFG

S → Xa

X → aX | bX | Λ

Old nullable Production	New Production
S → Xa	S → a
X → aX	X → a
X → bX	X → b

So the new CFG is

S → Xa | a

X → aX | bX | a | b

Example

S → XY
X → Zb
Y → bW
Z → AB
W → Z
A → aA | bA | Λ
B → Ba | Bb | Λ

- Null-able Non-terminals are?
- A, B, Z and W

S → XY
X → Zb
Y → bW
Z → AB
W → Z
A → aA | bA | Λ
B → Ba | Bb | Λ

Example Contd.

Old nullable Production	New Production
X → Zb	X → b
Y → bW	Y → b
Z → AB	Z → A and Z → B
W → Z	Nothing new
A → aA	A → a
A → bA	A → b
B → Ba	B →a
B → Bb	B → b

So the new CFG is

S → XY

X → Zb | b

Y → bW | b

Z → AB | A | B

W → Z

A → aA | bA | a | b

B → Ba | Ba | a | b

Killing unit-productions

- **Definition:** A production of the form
- Nonterminal → one Nonterminal

is called a **unit production**.

- The following theorem allows us to get rid of unit productions:

Theorem 24:

If there is a CFG for the language L that has no Λ-productions, then there is also a CFG for L with no Λ-productions and **no unit productions**.

Proof of Theorem 24

- This is another proof by constructive algorithm.
- **Algorithm:** For every pair of nonterminals A and B, if the CFG has a unit production A → B, or if there is a chain

  A → X_1 → X_2 → … → B

where X_1, X_2, ... are nonterminals, create new productions as follows:

- If the non-unit productions from B are

  B → s_1 | s_2| …

where s_1, s_2, ... are strings, we create the productions

A → s_1| s_2| …

Example

- Consider the CFG
  S → A| bb
  A → B | b
  B → S | a
- The non-unit productions are
  S → bb A → b B → a
- And unit productions are
  S → A
  A → B
  B → S

Example contd.

- Let's list all unit productions and their sequences and create new productions:

S → A	gives	S → b
S → A → B	gives	S → a
A → B	gives	A → a
A → B → S	gives	A → bb
B → S	gives	B → bb
B → S → A	gives	B → b

- Eliminating all unit productions, the new CFG is

  S → bb | b | a

  A → b | a | bb

  B → a | bb | b
- This CFG generates a finite language since there are no nonterminals in any strings produced from S.

Consider the CFG

S → A| bb

A → B | b

B → S | a

unit productions are

S → A

A → B

B → S

Useless Production

1. The productions that can never take part in derivation of any string. A – bc where a is not reachable from S (starting rule)

2. A variable (non-terminal) that can never take part in derivation of any string e.g. (B -> aB)

Another grammar:

$$S \rightarrow A$$

$$A \rightarrow aA$$

$$A \rightarrow \lambda$$

$$B \rightarrow bA$$ Useless Production

Not reachable from S

In general:

contains only terminals

if $S \Rightarrow \ldots \Rightarrow xAy \Rightarrow \ldots \Rightarrow w$

$w \in L(G)$

then variable A is useful

otherwise, variable A is useless

First: find all variables that can produce strings with only terminals

$S \rightarrow aS \mid A \mid C$

$A \rightarrow a$

$B \rightarrow aa$

$C \rightarrow aCb$

Round 1: $\{A, B\}$

$S \rightarrow A$

Round 2: $\{A, B, S\}$

Keep only the variables
that produce terminal symbols: $\{A, B, S\}$

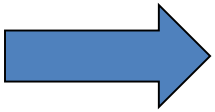
(the rest variables are useless)

$S \rightarrow aS \mid A \mid C$

$A \rightarrow a$

$B \rightarrow aa$

$C \rightarrow aCb$



$S \rightarrow aS \mid A$

$A \rightarrow a$

$B \rightarrow aa$

Remove useless productions

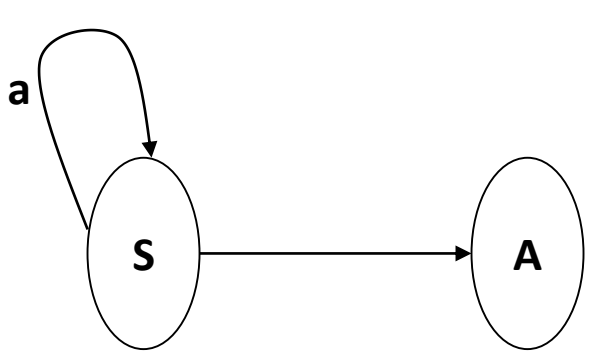
Second: Find all variables reachable from S

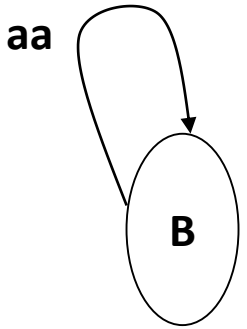
$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

Use a Dependency Graph





not reachable

Keep only the variables reachable from S

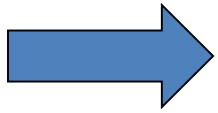
(the rest variables are useless)

Final Grammar

$S \rightarrow aS \mid A$

$A \rightarrow a$

~~$B \rightarrow aa$~~



$S \rightarrow aS \mid A$

$A \rightarrow a$

Remove useless productions

Normal Form of Context Free Grammars

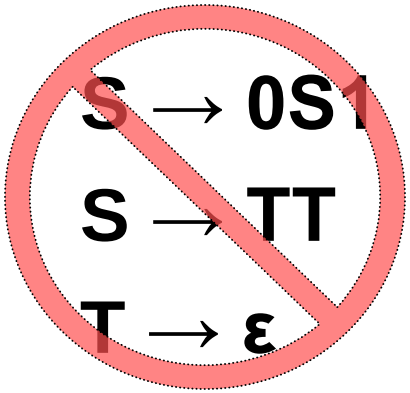
Normal Form

- To convert a grammar in normal form means “standardizing the productions without changing the resulting language”.

THE CHOMSKY NORMAL FORM

A context-free grammar is in Chomsky normal form if every rule is of the form:

$A \rightarrow BC$	B and C are not start variable
$A \rightarrow a$	a is a terminal
$S \rightarrow \varepsilon$	S is the start variable





$T \rightarrow 0$

$U \rightarrow SV$

$S \rightarrow TU \mid \varepsilon$

$V \rightarrow 1$

- S → AS|a
- A →SA|b

Is in Chomsky Normal Form. The Grammar

- S → AS|ASS
- A →SA|aa
- Is not; both production S → AS|ASS and A→SA|aa violate the conditions of definition.

Theorem: Any context-free language can be generated by a context-free grammar in Chomsky normal form

“Can transform any CFG into Chomsky normal form”

Chomsky Normal Form

There are 5 steps to follow in order to transform a grammar into CNF:

1. Add the a new start variable S_0 and the production rule $S_0 \longrightarrow S$.
2. Eliminate the ε-rules.
3. Eliminate the unary productions $A \longrightarrow B$.
4. Add rules of the form $V_t \longrightarrow t$ for every terminal t and replace t with the variable V_t.
5. Transform the remaining of the rules to the form $A \longrightarrow BC$ (A, B, C variables).

1. Add a new start variable

- We have to make sure that the start variable doesn't occur to the right side of some rule.
- Thus, we add a new start variable S_0 and the rule $S_0 \longrightarrow S$, where S is the old start variable.

2. Eliminate ε-rules

- We have to eliminate all productions of the form A ⟶ ε, for A being any non-start variable.
- To do so we should remove the rule A ⟶ ε and replace every appearance of A with ε in all other rules.

3. Eliminate unary productions

- A unary production is a production of the form $A \longrightarrow B$ (with both A, B being variables).
- There should only be productions of the form $V_1 \longrightarrow V_2V_3$ involving variables, thus we have to eliminate unary productions.
- To do so, we replace B in $A \longrightarrow B$ with the right parts of the rules involving B in the left part.

4. Add $V_t \longrightarrow t$ and replace t with V_t

- There should only be rules of the form $A \longrightarrow t$ involving terminals, thus terminals should disappear from every other rule involving more than just one single literal.
- To do so, we add a new variable V_t for every terminal t and we replace every appearance of t with V_t , except those in rules of the form $A \longrightarrow t$.

5. Transform rules to A ⟶ BC

- All the rules involving only variables should be of the form A ⟶ BC. Thus we should take care of all the rules involving more than 2 variables in the right part
- For the rule V ⟶ $A_1A_2A_3…A_n$,we start reducing the size of the right part by replacing every two variables with one new variable (resulting in the creation of n-2 new variables).

5. Transform rules to A ⟶ BC

$$V \longrightarrow A_1 A_2 A_3 A_4 A_5 A_6 \ldots A_n$$

5. Transform rules to A ⟶ BC

$$V \longrightarrow B_1 A_3 A_4 A_5 A_6 \ldots A_n$$

$$B_1 \longrightarrow A_1 A_2$$

5. Transform rules to A ⟶ BC

$$V \longrightarrow B_2A_4A_5A_6 \ldots A_n$$

$$B_2 \longrightarrow B_1A_3$$

$$B_1 \longrightarrow A_1A_2$$

5. Transform rules to A ⟶ BC

$$V \longrightarrow B_3A_5A_6 \ldots A_n$$

$$B_3 \longrightarrow B_2A_4$$

$$B_2 \longrightarrow B_1A_3$$

$$B_1 \longrightarrow A_1A_2$$

5. Transform rules to A ⟶ BC

$$V \longrightarrow B_4 A_6 \ldots A_n$$

$$B_4 \longrightarrow B_3 A_5$$

$$B_3 \longrightarrow B_2 A_4$$

$$B_2 \longrightarrow B_1 A_3$$

$$B_1 \longrightarrow A_1 A_2$$

5. Transform rules to A ⟶ BC

$$V \longrightarrow B_{n-2}A_n$$

$$B_{n-2} \longrightarrow B_{n-3}A_{n-1}$$

$$\dots$$

$$B_4 \longrightarrow B_3A_5$$

$$B_3 \longrightarrow B_2A_4$$

$$B_2 \longrightarrow B_1A_3$$

$$B_1 \longrightarrow A_1A_2$$

Example

$S \longrightarrow CSC \mid B$

$C \longrightarrow 00 \mid \varepsilon$

$B \longrightarrow 01B \mid 1$

Example

1. Add new start variablle

$S_0 \longrightarrow S$

$S \longrightarrow CSC \mid B$

$C \longrightarrow 00 \mid \varepsilon$

$B \longrightarrow 01B \mid 1$

Example

2. Eliminate ε-moves

$S_0 \longrightarrow S$

$S \longrightarrow CSC \mid B$

$C \longrightarrow 00 \mid \varepsilon$

$B \longrightarrow 01B \mid 1$

Example

2. Eliminate ε-moves

$S_0 \longrightarrow S$

$S \longrightarrow CSC \mid B \mid CS \mid SC \mid S$

$C \longrightarrow 00$

$B \longrightarrow 01B \mid 1$

Example

3. Eliminate Unary Productions

$S_0 \longrightarrow S$

$S \longrightarrow CSC \mid B \mid CS \mid SC \mid \cancel{S}$

$C \longrightarrow 00$

$B \longrightarrow 01B \mid 1$

Example

3. Eliminate Unary Productions

$S_0 \longrightarrow S$

$S \longrightarrow CSC \mid B \mid CS \mid SC$

$C \longrightarrow 00$

$B \longrightarrow 01B \mid 1$

Example

3. Eliminate Unary Productions

$S_0 \longrightarrow S$

$S \longrightarrow CSC \mid ~~B~~ \mid CS \mid SC$

$C \longrightarrow 00$

$B \longrightarrow 01B \mid 1$

Example

3. Eliminate Unary Productions

$S_0 \longrightarrow S$

$S \longrightarrow CSC \mid 01B \mid 1 \mid CS \mid SC$

$C \longrightarrow 00$

$B \longrightarrow 01B \mid 1$

Example

3. Eliminate Unary Productions

$S_0 \longrightarrow \cancel{S}$

$S \longrightarrow CSC \mid 01B \mid 1 \mid CS \mid SC$

$C \longrightarrow 00$

$B \longrightarrow 01B \mid 1$

Example

3. Eliminate Unary Productions

$S_0 \longrightarrow CSC \mid 01B \mid 1 \mid CS \mid SC$

$S \longrightarrow CSC \mid 01B \mid 1 \mid CS \mid SC$

$C \longrightarrow 00$

$B \longrightarrow 01B \mid 1$

Example

4. Create V_t for every terminal t

$S_0 \longrightarrow CSC \mid 01B \mid 1 \mid CS \mid SC$

$S \longrightarrow CSC \mid 01B \mid 1 \mid CS \mid SC$

$C \longrightarrow 00$

$B \longrightarrow 01B \mid 1$

$Z \longrightarrow 0$

Example

4. Create V_t for every terminal t

$S_0 \longrightarrow CSC \mid Z1B \mid 1 \mid CS \mid SC$

$S \longrightarrow CSC \mid Z1B \mid 1 \mid CS \mid SC$

$C \longrightarrow ZZ$

$B \longrightarrow Z1B \mid 1$

$Z \longrightarrow 0$

Example

4. Create V_t for every terminal t

$S_0 \longrightarrow CSC \mid Z1B \mid 1 \mid CS \mid SC$

$S \longrightarrow CSC \mid Z1B \mid 1 \mid CS \mid SC$

$C \longrightarrow ZZ$

$B \longrightarrow Z1B \mid 1$

$Z \longrightarrow 0$

$A \longrightarrow 1$

Example

4. Create V_t for every terminal t

$S_0 \longrightarrow CSC \mid ZAB \mid 1 \mid CS \mid SC$

$S \longrightarrow CSC \mid ZAB \mid 1 \mid CS \mid SC$

$C \longrightarrow ZZ$

$B \longrightarrow ZAB \mid 1$

$Z \longrightarrow 0$

$A \longrightarrow 1$

Example

5. Take care of long rules

$S_0 \longrightarrow CSC \mid ZAB \mid 1 \mid CS \mid SC$

$S \longrightarrow CSC \mid ZAB \mid 1 \mid CS \mid SC$

$C \longrightarrow ZZ$

$B \longrightarrow ZAB \mid 1$

$Z \longrightarrow 0$

$A \longrightarrow 1$

$D \longrightarrow CS$

Example

5. Take care of long rules

$S_0 \longrightarrow DC \mid ZAB \mid 1 \mid CS \mid SC$

$S \longrightarrow DC \mid ZAB \mid 1 \mid CS \mid SC$

$C \longrightarrow ZZ$

$B \longrightarrow ZAB \mid 1$

$Z \longrightarrow 0$

$A \longrightarrow 1$

$D \longrightarrow CS$

Example

5. Take care of long rules

$S_0 \longrightarrow DC \mid ZAB \mid 1 \mid CS \mid SC$

$S \longrightarrow DC \mid ZAB \mid 1 \mid CS \mid SC$

$C \longrightarrow ZZ$

$B \longrightarrow ZAB \mid 1$

$Z \longrightarrow 0$

$A \longrightarrow 1$

$D \longrightarrow CS$

$E \longrightarrow ZA$

Example

5. Take care of long rules

$S_0 \longrightarrow DC \mid EB \mid 1 \mid CS \mid SC$

$S \longrightarrow DC \mid EB \mid 1 \mid CS \mid SC$

$C \longrightarrow ZZ$

$B \longrightarrow EB \mid 1$

$Z \longrightarrow 0$

$A \longrightarrow 1$

$D \longrightarrow CS$

$E \longrightarrow ZA$

Convert the following into Chomsky normal form:

$$A \rightarrow BAB \mid B \mid \varepsilon$$
$$B \rightarrow 00 \mid \varepsilon$$

$S_0 \rightarrow A$

$A \rightarrow BAB \mid B \mid \varepsilon$

$B \rightarrow 00 \mid \varepsilon$



$S_0 \rightarrow A \mid \varepsilon$

$A \rightarrow BAB \mid B \mid BB \mid AB \mid BA$

$B \rightarrow 00$



$S_0 \rightarrow BAB \mid 00 \mid BB \mid AB \mid BA \mid \varepsilon$

$A \rightarrow BAB \mid 00 \mid BB \mid AB \mid BA$

$B \rightarrow 00$



$S_0 \rightarrow BC \mid DD \mid BB \mid AB \mid BA \mid \varepsilon, \quad C \rightarrow AB,$

$A \rightarrow BC \mid DD \mid BB \mid AB \mid BA, \quad B \rightarrow DD, \quad D \rightarrow 0$